

BUAN301 – In-Class Exercise (Probability)**Question 1**

Suppose we have a sample space with five equally likely outcomes i.e. $S = \{E_1, E_2, E_3, E_4, E_5\}$. Let events A , B and C be defined as $A = \{E_1, E_2\}$, $B = \{E_3, E_4\}$ and $C = \{E_2, E_3, E_5\}$.

- a) Find $P(A)$, $P(B)$, $P(C)$
- b) Find $P(A \cup B)$. Are A and B mutually exclusive?
- c) Find $A \cup B^c$ and $P(A \cup B^c)$.
- d) Find $P(B \cup C)$

Question 2

At an e-commerce customer service centre, a total of 112 complaints were received over two issues. 78 customers complained about late delivery of the items and 40 complained about poor product quality.

- a) Calculate the probability that a customer will complain about both late delivery and product quality.
- b) What is the probability that a customer has complained only about poor quality of the product?

Question 3

A committee of 5 members is to be selected from a group of 6 men and 9 women randomly. What is the probability that the committee will have 3 men and 2 women?